CP2.2 — Series: Maclaurin & Taylor

1. Maclaurin Series

Maclaurin Series: If $f(x)$ is infinitely differentiable at $x = 0$, then:

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \cdots = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n$$

This is a power series expansion about $x = 0$. The series converges to $f(x)$ within the **radius of convergence**.

2. Standard Maclaurin Series (Must Know)

Function	Maclaurin Series	Radius of Convergence
e^x	$1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots = \sum_{n=0}^{\infty} \frac{x^n}{n!}$	All x
$\sin x$	$x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$	All x
$\cos x$	$1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$	All x
$\ln(1+x)$	$x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^n}{n}$	$-1 < x \leq 1$
$(1+x)^n$	$1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \cdots$	$\lvert x \rvert < 1$ (for non-integer n)

Exam tip: These are in the Formulae Booklet — but you must know how to use and manipulate them quickly. The general term pattern is essential for "state the general term" questions.

2.1 Deriving eˣ from First Principles

Let $f(x) = e^x$. Then $f^{(n)}(x) = e^x$ for all n, so $f^{(n)}(0) = 1$ for all n. Therefore:

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

2.2 Deriving sin x and cos x

For $f(x) = \sin x$: derivatives cycle as $\cos x, -\sin x, -\cos x, \sin x, \ldots$. Evaluating at 0: $0, 1, 0, -1, 0, 1, \ldots$, giving only odd-power terms with alternating signs.

3. Deriving Maclaurin Series from Scratch

Example: Find the first four non-zero terms of the Maclaurin series for $f(x) = \ln(1 + x)$.

$f(x) = \ln(1 + x)$, $f(0) = 0$
$f'(x) = (1 + x)^{-1}$, $f'(0) = 1$
$f''(x) = -(1 + x)^{-2}$, $f''(0) = -1$
$f'''(x) = 2(1 + x)^{-3}$, $f'''(0) = 2$
$f^{(4)}(x) = -6(1 + x)^{-4}$, $f^{(4)}(0) = -6$

$$\ln(1 + x) = 0 + x + \frac{-1}{2!}x^2 + \frac{2}{3!}x^3 + \frac{-6}{4!}x^4 + \cdots = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots$$

4. Combining and Composing Maclaurin Series

To find the series of a composite function, substitute one series into another, keeping terms up to the required power.

Example: Find the first four non-zero terms of $e^{\sin x}$.

We use $\sin x = x - \frac{x^3}{6} + \frac{x^5}{120} - \cdots$ and $e^u = 1 + u + \frac{u^2}{2} + \frac{u^3}{6} + \cdots$

Let $u = x - \frac{x^3}{6} + \cdots$ Then:

$$u^2 = \left(x - \frac{x^3}{6} + \cdots\right)^2 = x^2 - \frac{x^4}{3} + \cdots$$

$$u^3 = x^3 + \cdots \quad (\text{higher order terms negligible for } \leq x^4)$$

$$e^{\sin x} = 1 + \left(x - \frac{x^3}{6}\right) + \frac{1}{2}\left(x^2 - \frac{x^4}{3}\right) + \frac{x^3}{6} + \cdots$$

$$= 1 + x + \frac{x^2}{2} + 0 \cdot x^3 - \frac{x^4}{8} + \cdots$$

So: $e^{\sin x} = 1 + x + \frac{x^2}{2} - \frac{x^4}{8} + \cdots$

Example: Find the series for $\sin(x^2)$ up to the term in x^{10}.

Replace x with x^2 in $\sin x = x - \frac{x^3}{6} + \frac{x^5}{120} - \cdots$:

$$\sin(x^2) = x^2 - \frac{x^6}{6} + \frac{x^{10}}{120} - \cdots$$

Example: Find the first three non-zero terms of $\cos^2 x$.

Use $\cos 2x = 1 - 2\sin^2 x$, so $\cos^2 x = \frac{1 + \cos 2x}{2}$.

$\cos 2x = 1 - \frac{(2x)^2}{2!} + \frac{(2x)^4}{4!} - \cdots = 1 - 2x^2 + \frac{2x^4}{3} - \cdots$

$$\cos^2 x = \frac{1 + \cos 2x}{2} = 1 - x^2 + \frac{x^4}{3} - \cdots$$

Exam tip: When composing series, work systematically by power of x. Keep track of which terms can contribute to the highest power you need — discard terms that are definitely too high order early to save time.

5. Taylor Series

Taylor Series about $x = a$:

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3 + \cdots$$
$$= \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n$$

The Maclaurin series is the special case $a = 0$.

Alternative form — Taylor's theorem with $x = a + h$:

$$f(a+h) = f(a) + hf'(a) + \frac{h^2}{2!}f''(a) + \frac{h^3}{3!}f'''(a) + \cdots$$

5.1 Using Taylor Series to Approximate Values

Example: Find a Taylor expansion of $\ln x$ about $x = 1$, up to the term in $(x-1)^4$. Hence approximate $\ln(1.1)$.

Let $a = 1$. $f(x) = \ln x$, $f(1) = 0$
$f'(x) = 1/x$, $f'(1) = 1$
$f''(x) = -1/x^2$, $f''(1) = -1$
$f'''(x) = 2/x^3$, $f'''(1) = 2$
$f^{(4)}(x) = -6/x^4$, $f^{(4)}(1) = -6$

$$\ln x = (x-1) - \frac{(x-1)^2}{2} + \frac{(x-1)^3}{3} - \frac{(x-1)^4}{4} + \cdots$$

Setting $x = 1.1$ (so $x - 1 = 0.1$):

$$\ln(1.1) \approx 0.1 - 0.005 + 0.000\overline{3} - 0.000025 = 0.09531\ldots$$

Example: Expand $f(x) = \sin x$ about $x = \pi/6$, giving the first three non-zero terms.

$f(\pi/6) = 1/2$, $f'(\pi/6) = \cos(\pi/6) = \sqrt{3}/2$
$f''(\pi/6) = -\sin(\pi/6) = -1/2$, $f'''(\pi/6) = -\cos(\pi/6) = -\sqrt{3}/2$

$$\sin x = \frac{1}{2} + \frac{\sqrt{3}}{2}\left(x - \frac{\pi}{6}\right) - \frac{1}{4}\left(x - \frac{\pi}{6}\right)^2 - \frac{\sqrt{3}}{12}\left(x - \frac{\pi}{6}\right)^3 + \cdots$$

6. Finding Series by Differentiation or Integration

Sometimes it is easier to differentiate or integrate a known series to get a new one.

Example: Find the Maclaurin series for $\arctan x$.

Start from $\frac{d}{dx}(\arctan x) = \frac{1}{1+x^2}$.
Use the geometric series: $\frac{1}{1+x^2} = 1 - x^2 + x^4 - x^6 + \cdots$ for $|x| < 1$.
Integrate term by term (with $\arctan 0 = 0$):

$$\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1}, \quad |x| \le 1$$

Example: Find the series for $\dfrac{1}{(1-x)^2}$ using differentiation.

We know $\dfrac{1}{1-x} = 1 + x + x^2 + x^3 + \cdots$ for $|x| < 1$.

Differentiate: $\dfrac{1}{(1-x)^2} = 1 + 2x + 3x^2 + 4x^3 + \cdots = \displaystyle\sum_{n=0}^{\infty}(n+1)x^n$

7. Limits Using Series

Series provide an alternative to L'Hôpital's rule for evaluating limits of the form $\dfrac{0}{0}$.

Example: Evaluate $\displaystyle\lim_{x\to 0}\frac{\sin x - x}{x^3}$.

$$\sin x = x - \frac{x^3}{6} + \frac{x^5}{120} - \cdots$$
$$\sin x - x = -\frac{x^3}{6} + \frac{x^5}{120} - \cdots$$
$$\frac{\sin x - x}{x^3} = -\frac{1}{6} + \frac{x^2}{120} - \cdots \to -\frac{1}{6} \text{ as } x \to 0.$$

Example: Evaluate $\displaystyle\lim_{x\to 0}\frac{e^x - 1 - x - x^2/2}{x^3}$.

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \cdots$$
$$e^x - 1 - x - \frac{x^2}{2} = \frac{x^3}{6} + \frac{x^4}{24} + \cdots$$
$$\text{Limit} = \frac{1}{6}.$$

Exam tip: Using series for limits is often faster than L'Hôpital. Expand both numerator and denominator, then divide — the limit is the coefficient of the lowest power in the numerator divided by the lowest power in the denominator (if they match).

8. Radius of Convergence

Series	Converges for	Note
e^x	All $x \in \mathbb{R}$	Entire function
$\sin x$, $\cos x$	All $x \in \mathbb{R}$	Entire functions
$\ln(1+x)$	$-1 < x \leq 1$	Diverges for $x \leq -1$
$(1+x)^n$ ($n \notin \mathbb{Z}^+$)	$\lvert x \rvert < 1$	At endpoints depends on n
$\arctan x$	$\lvert x \rvert \leq 1$	Converges at $x = \pm 1$
$\frac{1}{1-x}$	$\lvert x \rvert < 1$	Geometric series

9. Working with Composite and Product Series

Example: Find the first four terms of the Maclaurin series for $e^x \cos x$.

$e^x = 1 + x + \dfrac{x^2}{2} + \dfrac{x^3}{6} + \cdots$
$\cos x = 1 - \dfrac{x^2}{2} + \dfrac{x^4}{24} - \cdots$
Multiply, collecting up to x^3:

$$e^x \cos x = \left(1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \cdots\right)\left(1 - \frac{x^2}{2} + \cdots\right)$$

Constant: 1
x^1: $1 \cdot x = x$
x^2: $\frac{x^2}{2} + 1 \cdot (-\frac{x^2}{2}) = 0$
x^3: $\frac{x^3}{6} + x \cdot (-\frac{x^2}{2}) = \frac{x^3}{6} - \frac{x^3}{2} = -\frac{x^3}{3}$

Therefore: $e^x \cos x = 1 + x - \dfrac{x^3}{3} + \cdots$

10. Key Technique — Building Series for "Show That" Questions

In "show that" questions you are given the answer and need to derive it step by step, showing all derivatives evaluated at 0 (or at a).

Example: Show that the Maclaurin series for $f(x) = \dfrac{1}{\sqrt{1-x}}$ begins $1 + \dfrac{x}{2} + \dfrac{3x^2}{8} + \cdots$

Using $(1+u)^n$ with $u = -x$, $n = -\frac{1}{2}$:

$$(1-x)^{-1/2} = 1 + \left(-\frac{1}{2}\right)(-x) + \frac{(-\frac{1}{2})(-\frac{3}{2})}{2!}(-x)^2 + \cdots$$

$$= 1 + \frac{x}{2} + \frac{\frac{3}{4}}{2}x^2 + \cdots = 1 + \frac{x}{2} + \frac{3x^2}{8} + \cdots \checkmark$$

11. Summary — Key Formulas

$$\text{Maclaurin: } f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$$

$$\text{Taylor about } a: f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n$$

12. Exam-Style Practice Questions

1. **[Maclaurin from scratch]** Find the first four non-zero terms of the Maclaurin series for $f(x) = e^{2x}\ln(1+x)$.
2. **[Composition]** Find the first three non-zero terms of the Maclaurin series for $f(x) = \cos(\sqrt{x})$, for $x \geq 0$. (Hint: replace x with $\sqrt{x}$ in the cosine series.)

3. **[Taylor]** Expand $f(x) = \tan x$ as a Taylor series about $x = \pi/4$, giving terms up to $(x - \pi/4)^2$.
4. **[Limits]** Use Maclaurin series to evaluate $\lim_{x \to 0} \dfrac{1 - \cos x}{x^2}$.
5. **[Limits]** Find $\lim_{x \to 0} \dfrac{\ln(1+x) - x + \frac{x^2}{2}}{x^3}$.
6. **[General term]** Write down the general term of the Maclaurin series for $\sinh x = \dfrac{e^x - e^{-x}}{2}$. State the radius of convergence.
7. **[Approximation]** Use the first three terms of an appropriate series to estimate $\sqrt{e}$. Compare with the exact value.

General exam tips for CP2.2:

- Always state what you are substituting and into which standard series
- For composite series, work term-by-term up to the required power — don't go further
- If asked "state the range of validity," use the table above
- For limits via series: the answer is the leading coefficient after cancellation with the denominator
- Check odd/even functions: $\sin x$, $\sinh x$ — odd, so only odd powers; $\cos x$, $\cosh x$ — even, only even powers